

TOSA Adjacent Channel Spacing: Problem Statement and Strategy Comparison

This report summarizes the problem of **TOSA (Tunable Optical Sub-Assembly) yield** with respect to **adjacent channel spacing error**, and compares three assembly strategies. The goal is to maximize the fraction of TOSAs that meet a **maximum absolute spacing error** threshold (e.g. ≤ 40 GHz).

1. Problem Statement

1.1 Context

- A **TOSA** contains **4 lasers**, one per channel (ch0–ch3), in a band-muxed architecture.
- Channel centers are defined in **frequency deviation** from a reference wavelength (1311 nm).
- For the **200 GHz class** (the case analyzed here), the four channel centers are:
 - ch0**: 0 GHz
 - ch1**: +200 GHz
 - ch2**: –200 GHz
 - ch3**: –400 GHz(all relative to the frequency at 1311 nm).
- Each laser’s actual frequency varies around its channel center due to manufacturing spread. We assume a **Gaussian** distribution with $3\sigma = \pm 1$ nm in wavelength at 1311 nm, which corresponds to $\sigma \approx 58.2$ GHz in frequency.

Laser frequency distributions by channel (200 GHz and 400 GHz spacing, centered at 1311 nm):

 Laser frequency histograms by channel

Reference wavelength distributions (4 lasers at 1311 nm, $3\sigma = \pm 1$ nm):

 Laser wavelength distributions

1.2 Adjacent Spacing Error

- Nominal adjacent spacings** (target values) are:
 - ch1 – ch0: **200 GHz**
 - ch2 – ch1: **–400 GHz**
 - ch3 – ch2: **–200 GHz**
- For a given TOSA (four laser frequencies), we compute the **three adjacent spacing errors**:
 - Spacing error (ch0–ch1) = $(f_1 - f_0) - 200$
 - Spacing error (ch1–ch2) = $(f_2 - f_1) - (-400)$
 - Spacing error (ch2–ch3) = $(f_3 - f_2) - (-200)$
- We care about the **maximum absolute spacing error** over these three pairs **per TOSA**:
 - Max |error| per TOSA** = $\max\{|e_{01}|, |e_{12}|, |e_{23}|\}$ (in GHz).

1.3 Yield Metric

- A TOSA is considered **in-spec** for a threshold **T** (e.g. 40 GHz) if:
 - Max |spacing error| $\leq T$ GHz.**
 - Yield** = (number of TOSAs with max |error| $\leq T$) / (total number of TOSAs built), expressed as a percentage.
 - The report focuses on **maximizing yield at ≤ 40 GHz** and also reports yields at 20, 100, and 200 GHz for context. At 1311 nm, **40 GHz $\approx \pm 0.23$ nm** wavelength error.
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2. Strategies Compared

Three strategies are implemented in this folder. All use the same channel grid and laser statistics (Gaussian, $3\sigma = \pm 1$ nm at 1311 nm).

2.1 Random Sampling

- Method**: For each TOSA, **randomly** pick one laser from each channel (independent Gaussian samples around each channel center).
- No binning or matching** across channels.
- Script**: `random_sampling_tosa.py`
- Sample size (this report)**: 10 M TOSAs (200 GHz class), seed 42.

2.2 Binned Sampling — 4-Bin Scheme

- Method**:

- Sample a **full inventory** of lasers per channel (**10 M per channel**).
- Assign each laser to one of **4 bins** by its **deviation from target** (GHz):
 - **BIN0**: -1000 to -100
 - **BIN1**: -100 to 0
 - **BIN2**: 0 to 100
 - **BIN3**: 100 to 1000
- Build TOSAs by selecting **one laser per channel from the same bin index** (e.g. one from BIN2 for ch0, one from BIN2 for ch1, etc.).
- **No per-bin cap**: for each bin we build as many TOSAs as the inventory allows (minimum across the four channel-bin counts). Total TOSAs ≈ 10 M.
- **Script**: `binned_sampling_tosa.py`
- **Total TOSAs (this report)**: ~10 M (full inventory).

2.3 Binned Sampling — 10-Bin Scheme

- **Method**: Same idea as 4-bin, but with **10 bins of 40 GHz width** in deviation from target:
 - BIN0: -1000 to -160, BIN1: -160 to -120, ..., BIN9: 160 to 200 GHz.
 - TOSAs are again built by picking one laser per channel from the **same bin index**. Full inventory per bin (no cap).
- **Script**: `binned_sampling_tosa.py`
- **Total TOSAs (this report)**: ~10 M (full inventory).

2.4 Bin definitions (4-bin and 10-bin)

The figure below shows how the **4-bin** and **10-bin** schemes partition laser **deviation from target** (GHz). The gray curve is the laser deviation PDF (Gaussian, $\sigma \approx 58.2$ GHz). Colored vertical bands are the bins; TOSAs are built by picking one laser per channel from the same bin.

Bin definitions: 4-bin and 10-bin schemes

3. Yield Comparison (Seed 42)

Numerical results from the scripts in this folder (200 GHz class only):

Max error threshold	Random (% TOSAs)	4-bin (% TOSAs)	10-bin (% TOSAs)	Wavelength error (nm)
≤ 20 GHz	0.98	9.37	47.48	±0.115
≤ 40 GHz	6.87	40.30	99.88	±0.229
≤ 100 GHz	51.39	99.89	100.00	±0.573
≤ 200 GHz	95.86	100.00	100.00	±1.147

TOSA counts (this run):

- Random: 10,000,000
 - 4-bin: 9,995,164
 - 10-bin: 9,993,758
- All three strategies use comparable sample sizes (~10 M TOSAs). Binned sampling uses a **full inventory of 10 M lasers per channel** with **no per-bin cap**: for each bin we build as many TOSAs as the smallest channel-bin count allows, so nearly the entire inventory is converted to TOSAs.

Random sampling — adjacent spacing error histograms and yield table:

Random sampling: spacing error (signed), |error|, max |error| PDF, max |error| CDF

Binned sampling (4-bin and 10-bin) — adjacent spacing error histograms and yield table:

Binned sampling: 4-bin and 10-bin spacing error panels and table

4. Which Strategy Gives Maximum Yield for ≤ 40 GHz?

- **Random**: **6.87%** of TOSAs have max |spacing error| ≤ 40 GHz.
- **4-bin binned**: **40.30%** — much better than random.
- **10-bin binned**: **99.88%** — almost all TOSAs meet the 40 GHz spec.

Conclusion: The 10-bin binned strategy gives the best yield for ≤ 40 GHz |error| in this setup. By grouping lasers into narrow deviation bins and assembling TOSAs from the same bin index (full inventory, no per-bin cap), adjacent channel spacing errors are greatly reduced, and the 10-bin scheme (40 GHz bin width) aligns well with the 40 GHz error target.

5. 8-Channel TOSA (200 GHz spacing)

The same process is repeated for an **8-channel TOSA** with **all adjacent channels spaced by 200 GHz**. Channel centers (frequency deviation from 1311 nm ref) are **0, 200, 400, 600, 800, 1000, 1200, 1400 GHz** (ch0–ch7). There are **7 adjacent spacing pairs** (ch0–ch1 through ch6–ch7); each has nominal spacing **200 GHz**. Max |error| per TOSA is the maximum of the 7 absolute spacing errors. Same laser statistics (Gaussian, $3\sigma = \pm 1$ nm at 1311 nm) and same strategies: **random sampling**, **4-bin binned**, and **10-bin binned** (same bin definitions by deviation from target; full inventory, no per-bin cap). Scripts: `random_sampling_tosa_8ch.py`, `binned_sampling_tosa_8ch.py`. Separate figure files are generated so 4-channel and 8-channel results are not overwritten.

5.1 Yield comparison (8-channel, seed 42)

Max error threshold	Random (% TOSAs)	4-bin (% TOSAs)	10-bin (% TOSAs)	Wavelength error (nm)
≤ 20 GHz	0.00	0.74	18.55	±0.115
≤ 40 GHz	0.27	13.85	99.77	±0.229
≤ 100 GHz	22.66	99.77	100.00	±0.573
≤ 200 GHz	90.79	100.00	100.00	±1.147

TOSA counts (8-channel run): Random: 10,000,000 | 4-bin: 9,992,829 | 10-bin: 9,989,597.

With more adjacent pairs (7 instead of 3), **random yield at ≤ 40 GHz drops sharply** (0.27% vs 6.87% for 4-channel). **10-bin binned again gives the best yield** (99.77% at ≤ 40 GHz); 4-bin is much better than random (13.85%) but well below 10-bin.

8-channel random sampling — adjacent spacing error histograms and yield table:

 8-channel random sampling: spacing error histograms and table

8-channel binned sampling (4-bin and 10-bin) — adjacent spacing error histograms and yield table:

 8-channel binned sampling: 4-bin and 10-bin spacing error panels and table

6. Artifacts in This Folder

File	Description
<code>random_sampling_tosa.py</code>	Random TOSA assembly; computes spacing errors and plots histograms + summary table.
<code>random_sampling_tosa_adjacent_spacing_error_histograms.png</code>	One row: signed spacing error, error , max error PDF, max error CDF; table with % TOSAs and λ error.
<code>binned_sampling_tosa.py</code>	4-bin and 10-bin binned TOSA assembly; same plot layout per scheme.
<code>binned_sampling_tosa_spacing_error_histograms.png</code>	Two rows (4-bin, 10-bin), four columns each (same as above); table with 4-bin and 10-bin % TOSAs and wavelength error.
<code>bin_definition_histogram.py</code>	Plots 4-bin and 10-bin scheme definitions (deviation from target) with laser PDF.
<code>bin_definition_histogram.png</code>	Single figure: two panels showing bin edges and laser deviation PDF for 4-bin and 10-bin.
<code>laser_frequency_histograms.py</code>	Histograms of laser frequency deviation per channel (200 / 400 GHz classes).
<code>laser_wavelength_distributions.py</code>	Wavelength distribution visualization (reference).
<code>random_sampling_tosa_8ch.py</code>	Random TOSA assembly for 8-channel 200 GHz; outputs <code>random_sampling_tosa_8ch_adjacent_spacing_error_histograms.png</code>
<code>random_sampling_tosa_8ch_adjacent_spacing_error_histograms.png</code>	8-channel random: signed error , max error PDF/CDF and table.
<code>binned_sampling_tosa_8ch.py</code>	Binned TOSA assembly for 8-channel 200 GHz; outputs <code>binned_sampling_tosa_8ch_spacing_error_histograms.png</code> .
<code>binned_sampling_tosa_8ch_spacing_error_histograms.png</code>	8-channel binned (4-bin and 10-bin) spacing error panels and table.

7. Reproducibility

- **Random:** `np.random.seed(42)` then `build_tosas_for_class(CHANNELS_200_GHZ, N_TOSAS_PER_CLASS)`.
- **Binned:** `np.random.default_rng(42)` then `run_binned_sampling_200ghz(rng)`. Uses 10 M lasers per channel and full inventory (no per-bin cap).
- Run `python random_sampling_tosa.py`, `python binned_sampling_tosa.py`, and `python bin_definition_histogram.py` from `architecture_band_muxed` to regenerate 4-channel figures. For 8-channel: `python random_sampling_tosa_8ch.py`, `python binned_sampling_tosa_8ch.py`.

Export to PDF (with figures)

To export this report to PDF with all figures embedded:

1. **Pandoc (command line)**
From the **repository root** (so image paths in the doc resolve correctly):

```
pandoc src/probable-potato-guide3-arch/architecture_band_muxed/TOSA_SPACING_YIELD_REPORT.md -o src/probable-potato-guide3-arch/architecture_band_muxed/TOSA_SPACING_YIELD_REPORT.pdf
```

Requires [pandoc](#) and a LaTeX engine (e.g. MacTeX, TeX Live) or use `--pdf-engine=wkhtmltopdf` if installed.

2. Cursor / VS Code

Install the **Markdown PDF** extension. Open this file, then run **Markdown PDF: Export (pdf)** from the Command Palette (**Cmd+Shift+P**). Export uses the same folder as the **.md** file; ensure image paths in the report match how the extension resolves paths (workspace root vs file-relative).

3. Jupyter notebook

Use **TOSA_SPACING_YIELD_REPORT.ipynb**: run all cells to generate figures, then **File → Print → Save as PDF** or **jupyter nbconvert --to pdf TOSA_SPACING_YIELD_REPORT.ipynb** (requires LaTeX or **--to webpdf** with playwright).